

Project 7/8

Camera settings research

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Contents

Summary	1
Prologue	1
Theoretical framework	2
How does camera resolution affect our line detection?	2
What is camera resolution?	2
What effect can resolution have on a video?	2
Method	3
How does camera frame rate affect our line detection?	4
What is camera frame rate?	4
What effect can frame rate have on a video?	4
Method	5
Results	6
Conclusion	6
Changelog	7

Summary

In order to have a functional line detection you need a good camera, and with a good camera comes camera setting. Within this research paper will be two experiments about those settings.

Experiment one: camera resolution.

After testing all the camera resolutions and putting those videos through the code the results were that 480 resolution is too grainy to use as our camera resolution. 1080 resolution is very sharp but it requires too much computing power in the long run. That only leave 720 as the best option, because it is sharp enough to not be grainy and it uses less computing power than 1080. My advice would be to use 720 resolution.

Experiment two: camera frame rate.

The given cameras from the RDW should be able to record at 30fps and at 60fps. But while testing this the results came that the camera could only record at 30fps. This immediately concludes my research, with the results that 30fps is the better choice.

Prologue

One of the challenges in the self-driving challenge is to detect white lines on the track and drive in between them. Our solution to this problem is to convert the images from the two outer cameras to black and white and apply Canny edge detection. This ensures clear white lines around all of the objects in the frame. In this research paper, experiments will be

conducted to investigate how camera resolution and frame rate affect the quality of line detection.

The main questions for this research paper are:

1. How does camera resolution affect our line detection?
2. How does camera frame rate affect our line detection?

Theoretical framework

How does camera resolution affect our line detection?

What is camera resolution?

Camera resolution refers the number of pixels in a camera frame. Pixels are tiny dots that form a digital image. The number of pixels determines how detailed and clear the image appears. The more pixels in an image the clearer the actual image is.

For example:

This is an image of an apple with the resolution of 225 x 225 pixels.	
This is the same image of the apple but than with the resolution of 90 x 90 pixels.	

The following attachments were used in writing this text:

Attachment 1.

What effect can resolution have on a video?

As stated previously, a higher resolution means that your video contains more pixels. With more pixels available the video itself will be more detailed, which makes it appear sharper and clearer. Lesser resolutions could maybe look blurry and unfocused but it also take up less memory and storage space.

Method

Type of research: experimental.

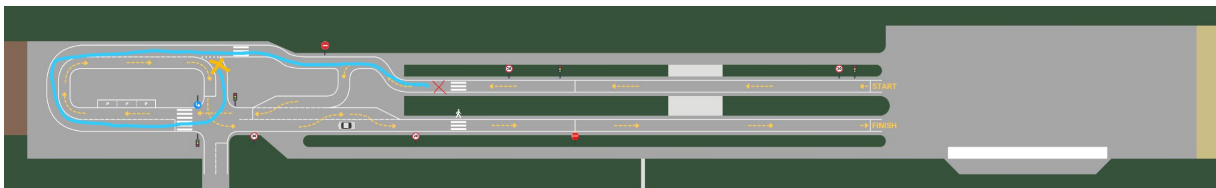
The results of the following experiments will help answer the main research questions.

Question: How does camera resolution affect line detection?

Why this experiment:

Supplies:

1. Cube NUC AI+, with the drivers for the controller already installed (see attachment 2)
2. HDMI cable
3. Computer screen with a HDMI port
4. Electricity
5. Keyboard and mouse, NOT Bluetooth
6. videoRecorder.py (see attachment 3)
7. conf.yaml (see attachment 4)
8. Kart, in the same setting as SDC closed challenge 2026
9. X-box controller
10. USB a -> USB c cable
11. Driver for the kart
12. Laptop with anydesk
13. Main1.py (see attachment 6)
14. lineDetector.py (see attachment 7)
15. LineThread.py (see attachment 8)



What is the experiment:

The goal of this experiment is to gather mostly identical video footage from the two outer cameras of the kart while using different camera frame rates. To achieve this, follow these

Hypothesis:

The line detection get sharper because the details are being seen faster. By that logic 1080 will be the resolution that will be used.

Procedure:

steps:

1. Connect the computer screen with the HDMI cable to the NUC.
2. Connect the keyboard and mouse to the NUC.
3. Supply power to the NUC and the screen and turn both devices on.
4. Copy both videoRecorder.py and conf.yaml to the NUC

5. Disconnect the NUC from the computer screen.
6. Plug the NUC into the kart.
7. Turn on the kart by pressing the on button.
8. Turn on the laptop.
9. Open the application anydesk.
10. Connect to the NUC through its IP address.
11. Go to the folder where you copied both of the code files.
12. Open a terminal here.
13. Make sure the kart stands on the red x.
14. Run 'videoRecorder.py test1'.
15. Plug in the x-box controller through the USB-c cable, into the NUC.
16. Step into the kart.
17. Turn on the motor by releasing the red button on the back of the kart, and pressing the first button on the speed remote until you hear 2 clicks.
18. Drive over the drawn blue line to the yellow x.
19. When you arrive at the yellow x, turn of the motor by pressing the big red button on the back.
20. Step out of the kart.
21. Press CTRL+C in the terminal on the laptop.
22. Repeat steps 13 to 21, with 'videoRecorder.py test3' and videoRecorder.py test5'.
23. Turn off the kart by pressing the off button.
24. Plug the NUC into the computer screen again.

Now we are going to compare the video fotage from test1, test3 and test5.

25. Copy 'main1.py', 'lineDetector.py' en 'LineThread.py' in a new folder on the NUC.
26. Open main1.py in a text editor.
27. Scroll to the bottom of the file until you see 'videoPath'.
28. Change the name between "" symbols to the name of the folder where he videos with 480 resolution are located.
29. Save 'main1.py'
30. Open a terminal in the folder where 'main1.py', 'lineDetector.py' and 'LineThread.py' are located
31. Run 'main1.py'.
32. Study the footage.
33. Repeat steps 26 to 32 with 720 resolution and 1080 resolution.
34. Compare all the video footage.

Results:

When all video footage is compared to one another, there are sustain things that stand out. One of those things is that 480 resolution is too little for our line detection. That is being shown in a video that is grainy and unreliable because it shows lines that are not there.

1080 resolution is sharp and it shows the lines quite good, but it requires too much computing power to use this in the long run.

720 resolution shows the lines quite sharp and there is little interference from light, the video footage is not grainy and it uses less computing power than 1080 resolution.

How does camera frame rate affect our line detection?

What is camera frame rate?

Camera frame rate refers to the number of frames captured per second. The frame rates 30 fps (frames per second) and 60 fps are discussed in this research paper. For instance, a frame rate of 30 fps means that every second the camera take 30 pictures and plays them after each other to create a second of film. Those pictures are called frames. In the same way 60 fps means that the camera take 60 frames very second.

What effect can frame rate have on a video?

The more frames you have every second the smoother the image is. Within the scope of this project the camera's are being used for line detection and object detection. Both of these topics require a camera that can capture enough frames to ensure a smooth motion and reduce blur. Bur is created by a low frame rate.

Method

Type of research: experimental.

The results of the following experiments will help answer the main research questions.

Question: How does camera frame rate affect our line detection?

Why this experiment:

This experiment is performed to determine what camera frame rate give the best results for our line detection.

Supplies:

1. Laptop with Linux OS
2. Camera, from the SDC kart 2026 (see attachment 5)
3. videoRecorder.py (see attachment 3)
4. Conf.yaml (see attachment 4)
5. Timer

What is the experiment:

The goal of this experiment is to gather mostly identical video footage from the two outer cameras of the kart while using different camera resolutions.

Before actually testing on the track, the code has been tested. While doing these test there has been discovered that the camera, that is installed on the kart, can not record in 60 fps even though on the camera it self states that 60 fps should be possible.

procedure:

With the writing of the test code, and testing it at school instead of at the testing location, I discovered that the camera could not handle 60 fps, even though the camera itself states it could. If you record with 30 fps the video is normal. If you record with 60 fps than the camera will still record 30 fps but it will still give 60 frames every second in the video. That means that the videos will always be at 2 two times speed.

To do:

1. Turn on the Linux laptop.
2. Plug in the camera.
3. Download both of the code files and put them in the same folder.

4. Open a terminal in that folder.
5. Run "videoRecorder.py test1"
6. Start your timer and wait for 30 seconds.
7. Press CTRL+C on your keyboard to end the code.
8. Repeat step 5,6 and 7 with test2, test3, test4, test5 and test6.

Results:

Code: videoRecorder.py

	1920x1080	1280x720	640x480
30	30 seconds	30 seconds	30 seconds
60	15 seconds	15 seconds	15 seconds

All the videos that recorded in 60fps are 1/2 times shorter than all the videos that were recorded in 30 fps. Recording in 60fps will only bring us negative effects because the videos will not be stored correctly. The better choice will be 30fps.

Results

The camera frame rate experiment shows that our camera is not able to record at 60fps. This shows in the following way:

	1920x1080	1280x720	640x480
30	30 seconds	30 seconds	30 seconds
60	15 seconds	15 seconds	15 seconds

Every video that is recorded in 60 fps halves itself in time, by only recording 30 frames but still playing 60 frames every second when you play the video.

The camera resolution experiment shows a good difference between all of the options. 480 resolution is too grainy and 1080 resolution give too much interference and needs too much computing power, therefor 720 is the ultimate resolution for us to use.

Conclusion

The goal of this research was to determine how camera resolution and frame rate can affect the quality of the line detection.

Based on the results of the experiments, camera frame rate can not be improved. The frame rate only functions on 30fps, so my advise is to use 30fps on all time.

The camera resolution on the other hand can be improved, within the restrictions of the challenge in 2026, my advise would be 720 resolution. Because you get the best video quality withing computing power of the NUC that is given. If in any future challenge the NUC would be replaced by something stronger, my advise would be to redo the experiment with 720 resolution and 1080 resolution. Because 1080 could bring some big improvements. For now 720 resolution is more than enough to work with.

Attachments

1. <https://www.unitxlabs.com/blog/camera-resolution-machine-vision-basics-2/>

2. https://azerty.nl/product/black-intel-soc-ultra-7-258v-integrated-graphics-n-a-32gbembedded-1tb-ssd-no-hdd-win-11-home-802-11ax-5-2-aircooling-120w-3y/9516612?utm_source=google&utm_medium=cpc&utm_campaign=&&utm_id=1973814685&utm_term=141722565502_&utm_content=645537961146&gad_source=1&gad_campaignid=1973814685&gclid=Cj0KCQjwve7NBhC-ARIsALZy9HVNztDJLIEg60o97GQgnelqMRgRPNMUCK2XAroVYepSjrMLTfDzUD4aAjWREALw_wcB
3. videoRecorder.py (see code folder)
4. Conf.yaml (see code folder)
5. https://www.logitech.com/nl-nl/shop/p/streamcam.960-001297?utm_source=Google&utm_medium=Paid-Search&utm_campaign=DEPT_FY26_Q3_NLD_LO_B2C_Always-on-PWS-Base-Plan_Google_CVR_na&gclid=Cj0KCQjw4PPNBhD8ARIsAMo-icwkMkvpQFIX383DBeBZNA2CpQyr7oj30Fb4070MNBOPZfniybx2urlaApEREALw_wcB
6. Main1.py (see code folder)
7. lineDetector.py (see code folder)
8. LineThread.py (see code folder)

Changelog

Date	Version	Changed
19/2	1	Which programming language is best for the kart, design made.
5/3	2	Research paper subject has been changed to camera resolution and frame rate. design has been made and prologue is written. A start has been made with theoretical framework and method.
10/3	2.1	Language rewrite
19/3	2.2	Experiments for camera resolution
20/3	2.3	Answered questions about frame rate
26/3	2.4	Rewrote dutch to english
17/6	2.5	
21/6	2.6	